MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, ADVANCED LOW-POWER SCHOTTKY TTL, DECODERS, MONOLITHIC SILICON

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. The specification covers the requirements for monolithic silicon, Advanced low-power Schottky TTL, decoder microcircuits. Two product assurance classes and a choice of case outline/lead finish are provided for each type and are reflected in the part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510, and as specified herein.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	Single 3 to 8 line decoder

1.2.2 Device class. The class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Letter	Case outline (see MIL-M-38510, appendix C)
E	D-2 (16-lead, 1/4" x 7/8") dual-in-line package
F	F-5 (16-lead, 1/4" x 3/8") flat package
X	C-2 (20 terminal, .350" x .350"), square chip carrier package,

1.3 Absolute maximum ratings.

Supply voltage range	-0.5 V to +7.0 V
Input voltage range	-1.5 V at -18 mA to +7.0 V
Storage temperature	-65°C to +150°C
Maximum power dissipation (P_D) per device 1/	
Device type 01	55.5 mW
Lead temperature (soldering, 10 seconds)	300°C
Thermal resistance, junction-to-case (θ_{JC}):	
Case E	50°C/W
Case F	70°C/W
Case X	60°C/W
Junction temperature (T_J)	175°C

1/ Must withstand the added P_D due to short circuit test (e.g., I_O).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Recommended operating conditions.

Supply voltage	4.5 V minimum to 5.5 V maximum
Minimum high level input voltage (V_{IH})	2.0 V
Maximum low level input voltage (V_{IL})	0.8 V
Case operating temperature range	-55°C to +125°C

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 herein.

3.2.1 Terminal connections. The terminal connections shall be as specified on figure 1.

3.2.2 Logic diagram and truth table. The logic diagram and truth table shall be as specified on figure 2.

3.2.3 Case outlines. The case outlines shall be as specified in 1.2.3.

3.2.4 Schematic circuits. The schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity as a prerequisite for qualification. All manufacturers' schematics shall be maintained and available upon request.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.5).

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$	Device type	Limits Min	Limits Max	Unit
High level output voltage	V_{OH}	V_{CC} = 4.5 V; I_{OH} = -400 µA V_{IL} = 0.8 V	01	2.5		V
Low level output voltage	V_{OL}	V_{CC} = 4.5 V; I_{OL} = 4 mA V_{IL} = 0.8 V; V_{IH} = 2.0 V	01		0.4	V
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V; T_C = +25°C I_{IN} = -18 mA	01		-1.5	V
Low level input current	I_{IL}	V_{CC} = 5.5 V V_{IN} = 0.4 V	01	0	-200	µA
High level input current	I_{IH1}	V_{CC} = 5.5 V V_{IN} = 2.7 V	01		20	µA
High level input current	I_{IH2}	V_{CC} = 5.5 V V_{IN} = 7.0 V	01		100	µA
Output current 1/	I_O	V_{CC} = 5.5 V V_O = 2.25 V	01	-30	-112	mA
Supply current	I_{CC}	V_{CC} = 5.5 V	01		10	mA
Propagation delay time low-to-high level A, B, or C to output	t_{PLH1}	V_{CC} = 5.0 V R_L = 500 Ω C_L = 50 pF ±10%	01	6	25	ns
Propagation delay time high-to-low level A, B, or C to output	t_{PHL1}			6	22	ns
Propagation delay time low-to-high level ENABLE to output	t_{PLH2}			4	20	ns
Propagation delay time high-to-low level ENABLE to output	t_{PHL2}			5	20	ns

1/ The output conditions have been chosen to produce a current that closely approximates one half of the true short-circuit output current, I_{OS}.

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3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the microcircuit, but shall be retained on the initial contract.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group 11 (see MIL-M-38510, apendix E).

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters (pre burn-in) (method 5004)	1	1
Final electrical test parameters (method 5004)	1*,2,3,7, 9,10,11	1*,2,3, 7,9
Group A test requirements (method 5005)	1,2,3,7, 8,9,10,11	1,2,3,7, 9,10,11
Group B electrical test requirements (method 5005)(subgroup 5)	1,2,3,7, 8,9,10,11	NA
Group C end-point electrical parameters (method 5005)	NA	1,2,3
Additional electrical subgroups for group C periodic inspections	NA	None
Group D end-point electrical parameters (method 5005)	1,2,3	1,2,3

*Indicates PDA applies to subgroup 1 (see 4.2c).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510, and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in test (method 1015 of MIL-STD-883).

(1) Test condition D or E, using the circuit shown on figure 3, or equivalent.

(2) T_A = +125°C, minimum.

b. Interim and final electrical tests shall be as specified in table II, except interim electrical tests prior to burn-in is optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) for class S devices shall be as specified in MIL-M-38510. The PDA for class B devices shall be 10 percent based on failures from group A, subgroup 1 tests after cooldown as the final electrical test in accordance with method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim electrical tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroup 1, after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

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4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510 and as specified herein. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 4, 5, and 6 of table I of method 5005 of MIL-STD-883 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883. Electrical tests shall be as specified in table II.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical tests shall be as specified in table II herein.

b. Steady state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D or E, using the circuit shown on figure 3, or equivalent.

(2) T_A = +125°C minimum.

(3) Test duration: 1,000 hours, except as permitted by appendix B of MIL-M-38510 and method 1005 of MIL-STD-883.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical tests shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be specified as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

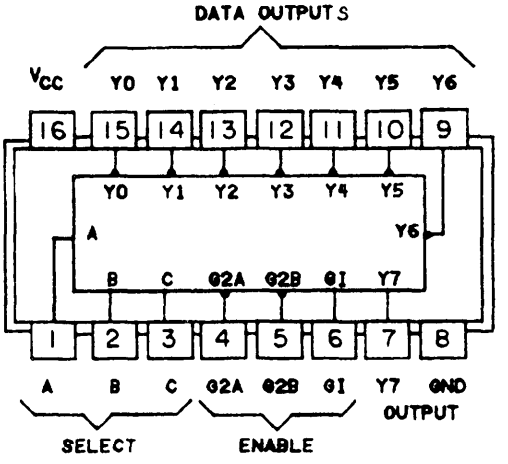
6. NOTES

6.1 Notes. The notes specified in MIL-M-38510 are applicable to this specification.

6.2 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logisitic support of existing equipment.

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Device type 01
Cases F or F



Device Type 01
Case X

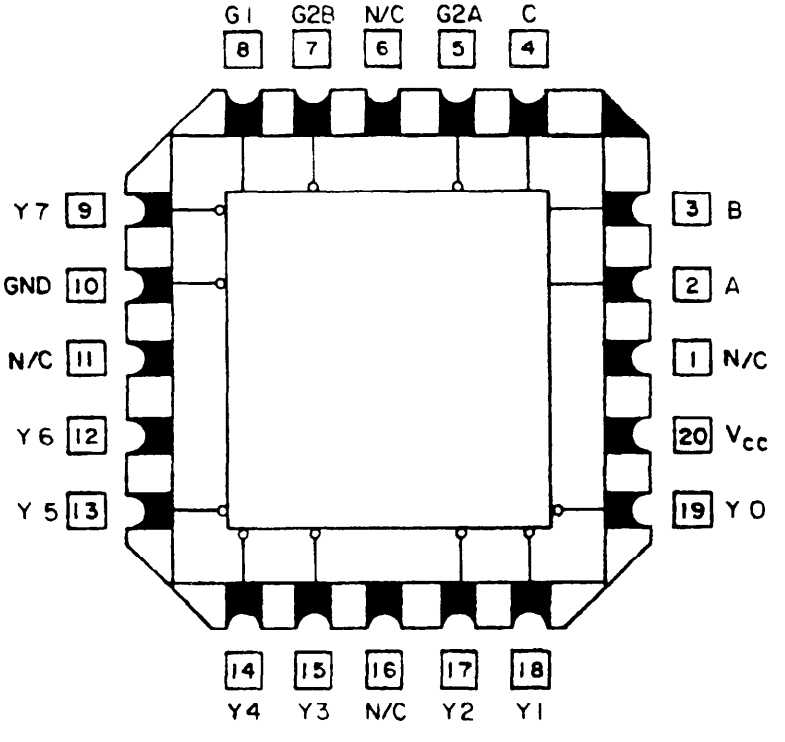
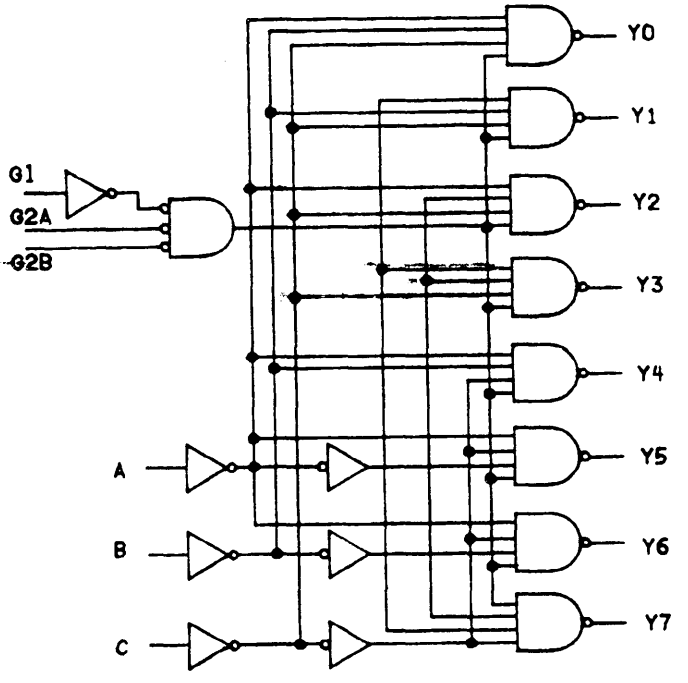


FIGURE 1. Terminal connections - top view.

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Device type 01



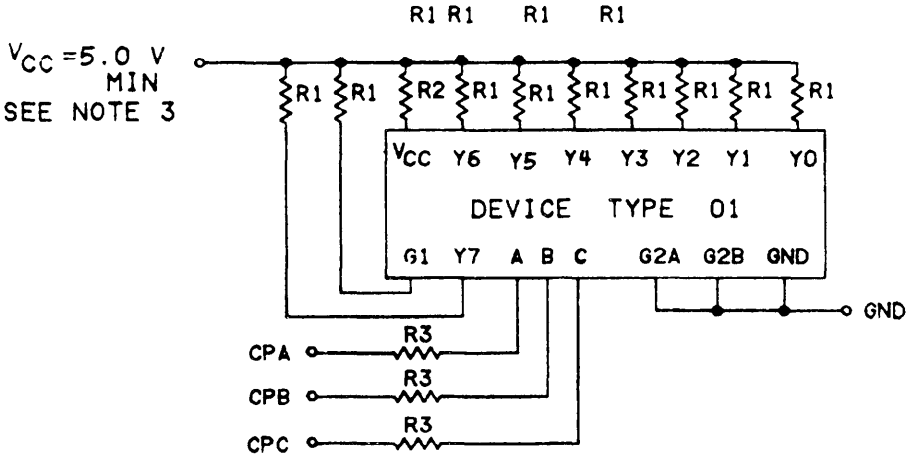
Device type 01

Inputs					Outputs							
Enable		Select										
G1	G2*	C	B	A	Y0	Y1	Y2	Y3	Y4	Y5	Y6	Y7
X	H	X	X	X	H	H	H	H	H	H	H	H
L	X	X	X	X	H	H	H	H	H	H	H	H
H	L	L	L	L	L	H	H	H	H	H	H	H
H	L	L	L	H	H	L	H	H	H	H	H	H
H	L	L	H	L	H	H	L	H	H	H	H	H
H	L	L	H	H	H	H	H	L	H	H	H	H
H	L	H	L	L	H	H	H	H	L	H	H	H
H	L	H	L	H	H	H	H	H	H	L	H	H
H	L	H	H	L	H	H	H	H	H	H	L	H
H	L	H	H	H	H	H	H	H	H	H	H	L

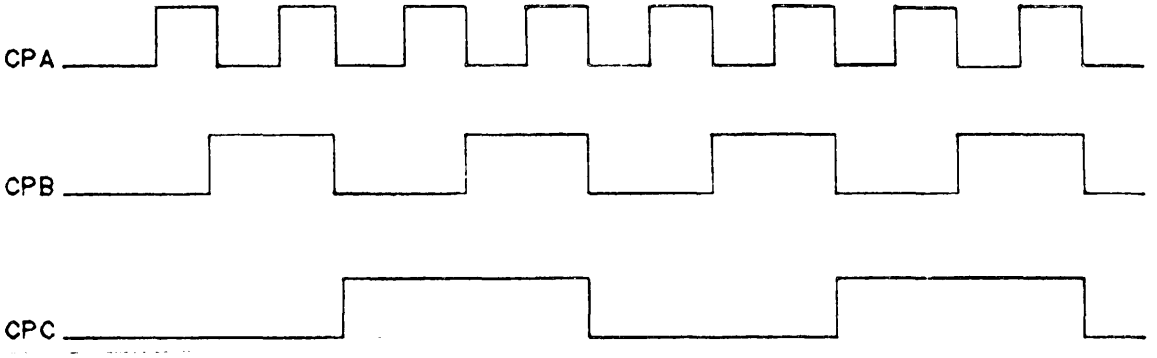
*G2 = G2A + G2B
H = high level, L = low level, X = irrelevant.

FIGURE 2. Logic diagram and truth table.

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NOTES:
1. CPA = 100 kHz ±50% square wave; duty cycle = 50% ±15%;
   V_{IL} = -0.5 V minimum to 0.8 maximum, V_{IH} = 2.0 V minimum to 5.5 V maximum.
   CPB = Same characteristics as CPA except 1/2 the frequency and phase relationship shown below.
   CPC = Same characteristics as CPB except 1/2 the frequency and phase relationship shown below.

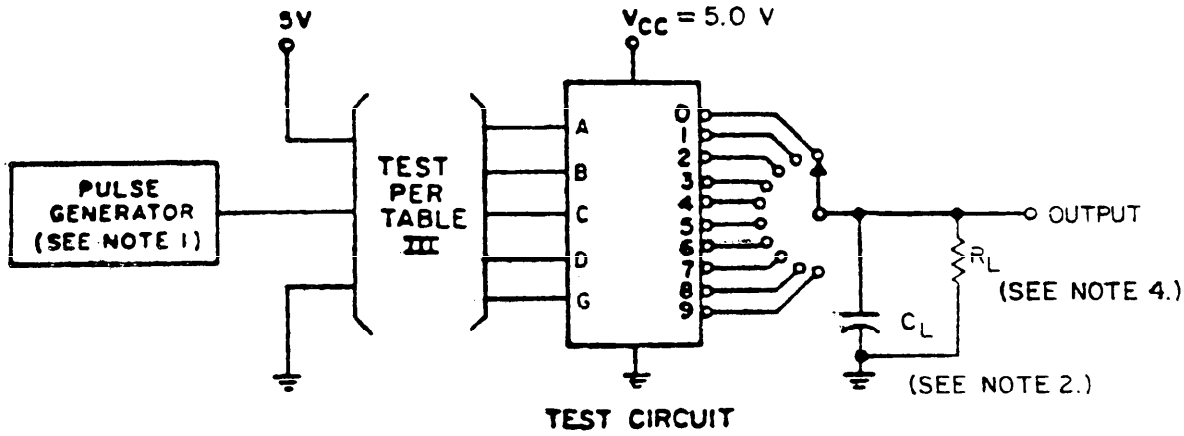


2. R1 = 1 kΩ ±5%; R3 = 27Ω ±5%.

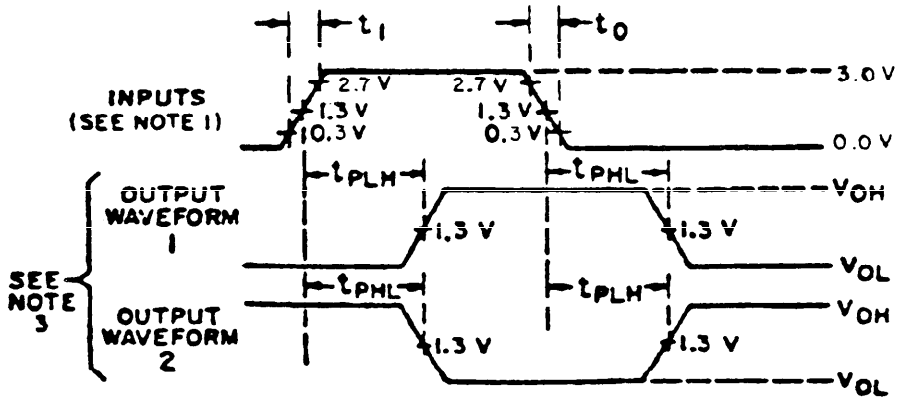
3. V_{CC} and R_2 shall be chosen to insure a 5.0 V minimum is present at device V_{CC} terminal.

FIGURE 3. Burn-in and life test circuits - Continued.

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TEST CIRCUIT



1. V_{IN} = Input pulse and has the following characteristics:

   PRR ≤ 1 MHz, t1 = t0 = 6.0 ±1.5 ns, Z_{OUT} ≅ 50Ω.

2. C_L = 50 pf ±10%, including scope probe, wiring and stray capacitance without package in test fixture.

3. Input-output waveform combination in accordance with the truth tables (see figure 2).

4. R_L = 499 ±1%

5. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 4. Switching times for device type 01.

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TABLE III. Group A inspection for device type 01.
Terminal conditions (pins not designated may high ≥2.0 V, or low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883-method	Case X 1/	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20	Measured terminal	Test limits		Unit
			Cases E,F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Min	Max	
			Test no.	A	B	C	G2A	G2B	G1	Y7	GND	Y6	Y5	Y4	Y3	Y2	Y1	Y0	V_{CC}				
1 T_C = 25°C	V_{OH}	3006	1						0.8 V		GND							-.4 mA	4.5 V	Y0	2.5		V
	"	"	2						"		"						-.4 mA	"	"	Y1	"		"
	"	"	3						"		"					-.4 mA	"	"	"	Y2	"		"
	"	"	4						"		"				-.4 mA	"	"	"	"	Y3	"		"
	"	"	5						"		"			-.4 mA	"	"	"	"	"	Y4	"		"
	"	"	6						"		"		-.4 mA	"	"	"	"	"	"	Y5	"		"
	"	"	7						"		"	-.4 mA	"	"	"	"	"	"	"	Y6	"		"
	"	"	8						"	-.4 mA	"	"	"	"	"	"	"	"	"	Y7	"		"
	V_{OL}	3007	9	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	2.0 V									4 mA	"	Y0		0.4	"
	"	"	10	2.0 V	0.8 V	"	"	"	"		"						4 mA	"	"	Y1		"	"
	"	"	11	0.8 V	2.0 V	2.0 V	"	"	"		"					4 mA	"	"	"	Y2		"	"
	"	"	12	0.8 V	2.0 V	0.8 V	"	"	"		"				4 mA	"	"	"	"	Y3		"	"
	"	"	13	2.0 V	0.8 V	0.8 V	"	"	"		"			4 mA	"	"	"	"	"	Y4		"	"
	"	"	14	"	0.8 V	2.0 V	"	"	"		"		4 mA	"	"	"	"	"	"	Y5		"	"
	"	"	15	"	2.0 V	0.8 V	"	"	"		"	4 mA	"	"	"	"	"	"	"	Y6		"	"
	"	"	16	"	2.0 V	2.0 V	"	"	"	4 mA	"	"	"	"	"	"	"	"	"	Y7		"	"
	V_{IC}		17	-18 mA															"	A		-1.5	"
			18		-18 mA														"	B		"	"
			19			-18 mA													"	C		"	"
			20				-18 mA												"	G2A		"	"
			21					-18 mA											"	G2B		"	"
			22						-18 mA										"	G1		"	"
	I_{IL}	3009	23	0.4 V															5.5 V	A	2/	2/	µA
	"	"	24		0.4 V														"	B	"	"	"
	"	"	25			0.4 V													"	C	"	"	"
	"	"	26				0.4 V												"	G2A	"	"	"
	"	"	27					0.4 V											"	G2B	"	"	"
	"	"	28						0.4 V										"	G1	"	"	"
	I_{IH1}	3010	29	2.7 V															"	A		20	"
	"	"	30		2.7 V														"	B		"	"
	"	"	31			2.7 V													"	C		"	"
	"	"	32				2.7 V												"	G2A		"	"
	"	"	33					2.7 V											"	G2B		"	"
	"	"	34						2.7 V										"	G1		"	"
	I_{IH2}	"	35	7.0 V															"	A		100	"
	"	"	36		7.0 V														"	B		"	"
	"	"	37			7.0 V													"	C		"	"
	"	"	38				7.0 V												"	G2A		"	"
	"	"	39					7.0 V											"	G2B		"	"
	"	"	40						7.0 V										"	G1		"	"
	I_O 3/	3011	41				5.5 V	5.5 V										2.25 V	"	Y0	-30 4/	-112 4/	mA
	"	"	42				"	"									2.25 V	"	"	Y1	"	"	"
	"	"	43				"	"								2.25 V	"	"	"	Y2	"	"	"
	"	"	44				"	"							2.25 V	"	"	"	"	Y3	"	"	"
	"	"	45				"	"						2.25 V	"	"	"	"	"	Y4	"	"	"
	"	"	46				"	"					2.25 V	"	"	"	"	"	"	Y5	"	"	"
	"	"	47				"	"				2.25 V	"	"	"	"	"	"	"	Y6	"	"	"
	"	"	48				"	"		2.25 V		"	"	"	"	"	"	"	"	Y7	"	"	"
	I_{CC}	3005	49						5.5 V										"	V_{CC}		10	mA
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																						

See [illegible]tes at end of table.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may high ≥2.0 V, or low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883-method	Case X 1/	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20	Measured terminal	Test limits		Unit
			Cases E,F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Min	Max	
			Test no.	A	B	C	G2A	G2B	G1	Y7	GND	Y6	Y5	Y4	Y3	Y2	Y1	Y0	V_{CC}				
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
7 5/ T_C = 25°C	Functional tests	3014	50	A	A	A	A	A	A	H	GND	H	H	H	H	H	H	H	5.0 V	All outputs	6/	6/	
			51	B	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"				
			52	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"				
			53	"	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"				
			54	"	"	"	B	"	"	"	"	"	"	"	"	"	"	"	"				
			55	"	"	"	A	B	"	"	"	"	"	"	"	"	"	"	"				
			56	"	"	"	B	A	"	"	"	"	"	"	"	"	"	"	"				
			57	"	"	"	"	B	"	"	"	"	"	"	"	"	"	"	"				
			58	"	"	"	"	"	A	"	"	"	"	"	"	"	"	"	"				
			59	"	"	"	"	"	B	"	"	"	"	"	"	"	"	"	"				
			60	A	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"				
			61	B	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"				
			62	A	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"				
			63	B	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"				
			64	A	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"				
			65	B	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"				
			66	A	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"				
8 5/	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and -55°C.																						
9 T_C = 25°C	t_{PLH1}	3003 (Fig. 4)	67	IN	GND	GND	GND	GND	5.0 V		GND								5.0 V	A to Y0	6	20	ns
			68	"	GND	"	"	"	"		"								"	A to Y1			
			69	"	5.0 V	"	"	"	"		"								"	A to Y2			
			70	"	5.0 V	"	"	"	"		"								"	A to Y3			
			71	"	GND	5.0 V	"	"	"		"								"	A to Y4			
			72	"	GND	"	"	"	"		"								"	A to Y5			
			73	"	5.0 V	"	"	"	"		"								"	A to Y6			
			74	"	5.0 V	"	"	"	"	OUT	"								"	A to Y7			
			75	GND	IN	GND	"	"	"		"								"	B to Y0			
			76	5.0 V	"	"	"	"	"		"								"	B to Y1			
			77	GND	"	"	"	"	"		"								"	B to Y2			
			78	5.0 V	"	"	"	"	"		"								"	B to Y3			
			79	GND	"	5.0 V	"	"	"		"								"	B to Y4			
			80	5.0 V	"	"	"	"	"		"								"	B to Y5			
			81	GND	"	"	"	"	"		"								"	B to Y6			
			82	5.0 V	"	"	"	"	"	OUT	"								"	B to Y7			
			83	GND	GND	IN	"	"	"		"								"	C to Y0			
			84	5.0 V	"	"	"	"	"		"								"	C to Y1			
			85	GND	5.0 V	"	"	"	"		"								"	C to Y2			
			86	5.0 V	5.0 V	"	"	"	"		"								"	C to Y3			
			87	GND	GND	"	"	"	"		"								"	C to Y4			
			88	5.0 V	GND	"	"	"	"		"								"	C to Y5			
			89	GND	5.0 V	"	"	"	"		"								"	C to Y6			
			90	5.0 V	5.0 V	"	"	"	"	OUT	"								"	C to Y7			

See footnotes at end of table.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may high ≥2.0 V, or low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883-method	Case X 1/	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20	Measured terminal	Test limits		Unit
			Cases E,F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16				
			Test no.	A	B	C	G2A	G2B	G1	Y7	GND	Y6	Y5	Y4	Y3	Y2	Y1	Y0	V_{CC}		Min	Max	
9 T_C = 25°C	t_{PHL1}	3003 (Fig. 4)	91	IN	GND	GND	GND	GND	5.0 V		GND							OUT	5.0 V	A to Y0	6	18	ns
			92	"	GND	"	"	"	"		"						OUT		"	A to Y1	"	"	"
			93	"	5.0 V	"	"	"	"		"					OUT			"	A to Y2	"	"	"
			94	"	5.0 V	"	"	"	"		"				OUT				"	A to Y3	"	"	"
			95	"	GND	5.0 V	"	"	"		"			OUT					"	A to Y4	"	"	"
			96	"	GND	"	"	"	"		"		OUT						"	A to Y5	"	"	"
			97	"	5.0 V	"	"	"	"		"	OUT							"	A to Y6	"	"	"
			98	"	5.0 V	"	"	"	"	OUT	"								"	A to Y7	"	"	"
			99	GND	IN	GND	"	"	"		"							OUT	"	B to Y0	"	"	"
			100	5.0 V	"	"	"	"	"		"						OUT		"	B to Y1	"	"	"
			101	GND	"	"	"	"	"		"					OUT			"	B to Y2	"	"	"
			102	5.0 V	"	"	"	"	"		"				OUT				"	B to Y3	"	"	"
			103	GND	"	5.0 V	"	"	"		"			OUT					"	B to Y4	"	"	"
			104	5.0 V	"	"	"	"	"		"		OUT						"	B to Y5	"	"	"
			105	GND	"	"	"	"	"		"	OUT							"	B to Y6	"	"	"
			106	5.0 V	"	"	"	"	"	OUT	"								"	B to Y7	"	"	"
			107	GND	GND	IN	"	"	"		"							OUT	"	C to Y0	"	"	"
			108	5.0 V	"	"	"	"	"		"						OUT		"	C to Y1	"	"	"
			109	GND	5.0 V	"	"	"	"		"					OUT			"	C to Y2	"	"	"
			110	5.0 V	"	"	"	"	"		"				OUT				"	C to Y3	"	"	"
			111	GND	GND	"	"	"	"		"			OUT					"	C to Y4	"	"	"
			112	5.0 V	GND	"	"	"	"		"		OUT						"	C to Y5	"	"	"
			113	GND	5.0 V	"	"	"	"		"	OUT							"	C to Y6	"	"	"
			114	5.0 V	5.0 V	"	"	"	"	OUT	"								"	C to Y7	"	"	"
	t_{PLH2}	"	115	GND	GND	GND	IN	"	"		"							OUT	"	G2A to Y0	4	17	"
		"	116	5.0 V	GND	"	"	"	"		"						OUT		"	G2A to Y1	"	"	"
		"	117	GND	5.0 V	"	"	"	"		"					OUT			"	G2A to Y2	"	"	"
		"	118	5.0 V	5.0 V	"	"	"	"		"				OUT				"	G2A to Y3	"	"	"
		"	119	GND	GND	5.0 V	"	"	"		"			OUT					"	G2A to Y4	"	"	"
		"	120	5.0 V	GND	"	"	"	"		"		OUT						"	G2A to Y5	"	"	"
		"	121	GND	5.0 V	"	"	"	"		"	OUT							"	G2A to Y6	"	"	"
		"	122	5.0 V	5.0 V	"	"	"	"	OUT	"								"	G2A to Y7	"	"	"
		"	123	GND	GND	GND	GND	IN	"		"							OUT	"	G2B to Y0	"	"	"
		"	124	5.0 V	GND	"	"	"	"		"						OUT		"	G2B to Y1	"	"	"
		"	125	GND	5.0 V	"	"	"	"		"					OUT			"	G2B to Y2	"	"	"
		"	126	5.0 V	5.0 V	"	"	"	"		"				OUT				"	G2B to Y3	"	"	"
		"	127	GND	GND	5.0 V	"	"	"		"			OUT					"	G2B to Y4	"	"	"
		"	128	5.0 V	GND	"	"	"	"		"		OUT						"	G2B to Y5	"	"	"
		"	129	GND	5.0 V	"	"	"	"		"	OUT							"	G2B to Y6	"	"	"
		"	130	5.0 V	5.0 V	"	"	"	"	OUT	"								"	G2B to Y7	"	"	"
		"	131	GND	GND	GND	"	GND	IN		"							OUT	"	G1 to Y0	"	"	"
		"	132	5.0 V	GND	"	"	"	"		"						OUT		"	G1 to Y1	"	"	"
		"	133	GND	5.0 V	"	"	"	"		"					OUT			"	G1 to Y2	"	"	"
		"	134	5.0 V	5.0 V	"	"	"	"		"				OUT				"	G1 to Y3	"	"	"
		"	135	GND	GND	5.0 V	"	"	"		"			OUT					"	G1 to Y4	"	"	"
		"	136	5.0 V	GND	"	"	"	"		"		OUT						"	G1 to Y5	"	"	"
		"	137	GND	5.0 V	"	"	"	"		"	OUT							"	G1 to Y6	"	"	"
	↓	"	138	5.0 V	5.0 V	"	"	"	"	OUT	"								"	G1 to Y7	"	"	"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may high ≥2.0 V, or low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Case X 1/	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
			Cases E,F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16				
			Test no.	A	B	C	G2A	G2B	G1	Y7	GND	Y6	Y5	Y4	Y3	Y2	Y1	YO	V_{CC}				
9 T_C = 25°C	t_{PHL2}	3003 (Fig. 4)	139	GND	GND	GND	IN	GND	5.0 V		GND							OUT	5.0 V	G2A to YO	5	17	ns
		"	140	5.0 V	GND	"	"	"	"		"					OUT	OUT		"	G2A to Y1	"	"	"
		"	141	GND	5.0 V	"	"	"	"		"				OUT				"	G2A to Y2	"	"	"
		"	142	5.0 V	5.0 V	"	"	"	"		"			OUT					"	G2A to Y3	"	"	"
		"	143	GND	GND	5.0 V	"	"	"		"		OUT						"	G2A to Y4	"	"	"
		"	144	5.0 V	GND	"	"	"	"		"	OUT							"	G2A to Y5	"	"	"
		"	145	GND	5.0 V	"	"	"	"		"								"	G2A to Y6	"	"	"
		"	146	5.0 V	5.0 V	"	"	"	"	OUT	"							OUT	"	G2A to Y7	"	"	"
		"	147	GND	GND	GND	GND	IN	"		"						OUT		"	G2B to YO	"	"	"
		"	148	5.0 V	GND	"	"	"	"		"					OUT			"	G2B to Y1	"	"	"
		"	149	GND	5.0 V	"	"	"	"		"				OUT				"	G2B to Y2	"	"	"
		"	150	5.0 V	5.0 V	"	"	"	"		"			OUT					"	G2B to Y3	"	"	"
		"	151	GND	GND	5.0 V	"	"	"		"		OUT						"	G2B to Y4	"	"	"
		"	152	5.0 V	GND	"	"	"	"		"	OUT							"	G2B to Y5	"	"	"
		"	153	GND	5.0 V	"	"	"	"		"								"	G2B to Y6	"	"	"
		"	154	5.0 V	5.0 V	"	"	"	"	OUT	"							OUT	"	G2B to Y7	"	"	"
		"	155	GND	GND	GND	"	GND	IN		"						OUT		"	G1 to YO	"	"	"
		"	156	5.0 V	GND	"	"	"	"		"					OUT			"	G1 to Y1	"	"	"
		"	157	GND	5.0 V	"	"	"	"		"				OUT				"	G1 to Y2	"	"	"
		"	158	5.0 V	5.0 V	"	"	"	"		"			OUT					"	G1 to Y3	"	"	"
		"	159	GND	GND	5.0 V	"	"	"		"		OUT						"	G1 to Y4	"	"	"
		"	160	5.0 V	GND	"	"	"	"		"	OUT							"	G1 to Y5	"	"	"
		"	161	GND	5.0 V	"	"	"	"		"								"	G1 to Y6	"	"	"
		"	162	5.0 V	5.0 V	"	"	"	"	OUT	"								"	G1 to Y7	"	"	"
10	t_{PLH1}	3003 (Fig. 4)	Same tests, terminal conditions as for subgroup 9, except T_C = +125°C																		6	25	"
	t_{PHL1}	"																			6	22	"
	t_{PLH2}	"																			4	20	"
	t_{PHL2}	"																			5	20	"
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																						

1/ For case X, pins not referenced are N/C.

2/ I_{IL} limits shall be as follows:

Test	Min/max limits in (μA) for circuit A	B	C
I_{IL}	0/-100	0/-100	0/-200

3/ Method 3011 shall be used, except the output voltage shall be as specified herein, and the output current shall be operating rather than short circuit current. The output conditions have been chosen to produce a current that closely approximates one half of the true short circuit output current I_{OS}.

4/ I_O limits for circuit "C" shall be -30/-110 mA.

5/ A ≥2.0 V and B ≤0.8 V.

6/ Output voltages shall be either:

(a) H = 2.5 V minimum and L = 0.4 V maximum when using a high speed checker double comparator or
(b) H ≥ 1.5 V and L ≤1.5 V when using a high speed checker single comparator.

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6.3 Ordering data. The contract or purchase order should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the qualify conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

h. Requirements for "JAN" marking.

6.4 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND - - - - - - - - - - - - - - - Electrical ground (common terminal)

I_{IN} - - - - - - - - - - - - - - - Current flowing into an input terminal.

V_{IN} - - - - - - - - - - - - - - - Voltage level at an input terminal.

6.5 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2) and lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.6 Substitutability. The cross-reference information below is pesented for the convenience of users. Microcircuits covered by this specification shall functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	54ALS138

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6.7 Manufacturers' designation. The manufacturer's circuits included in this specification are designated with an "X" as shown in table IV herein.

TABLE IV. Manufacturers' designation.

Device type	Circuits		
	A	B	C
	Texas Instruments	Motorola Inc.	National Semiconductor Corp
01	X		

Custodian:
Air Force - 17

Review activities:
Air Force - 11, 19, 85, 99
DLA - ES

Agent:
DLA - ES

Preparing activity:
Air Force - 17

(Project 5962-F615)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38510/377(USAF)	Microcircuits, Digital, Advanced Low-Power Schottky TTL, Decoders, Monolithic Silicon

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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